

FOCUSFLOW

Complete Project Documentation

An ADHD-oriented PDF learning workspace designed to make dense study material easier to enter, navigate, understand and sustain — while deliberately preventing the interface from becoming the distraction.

PROJECT	PURPOSE	CURRENT STACK
FocusFlow	Cognitive scaffolding around uploaded study material	HTML5 • CSS3 • ES6+ JavaScript • Python 3 http.server
Hackathon prototype	Reading + recall + regulation + organization	Modular JavaScript architecture + LocalStorage

This document consolidates the project description and implementation details supplied in the FocusFlow documentation. It preserves the project's terminology and distinguishes implemented features from the planned AI-assistant evolution.

1. The Problem

FocusFlow begins with a mismatch: a conventional PDF can contain excellent educational information while still presenting that information in a format that is difficult for some learners with ADHD to sustain attention on.

The project documentation specifically identifies challenges including working-memory overload, visual distraction, losing reading place, physical restlessness, context-switching fatigue, dense-text overwhelm, executive dysfunction and low motivation.

Why the problem compounds

FRICTION	WHAT HAPPENS	WHY IT MATTERS
Dense pages	Long passages and visually plain material demand sustained parsing	Attention is spread parsing the page instead of only processing the s
Losing place	A learner can lose the active line or paragraph	Time and working memory are spent reconstructing where to continue
Context switching	Notes, tasks and revision often live outside the viewing	Moving between tools adds another cognitive step.
Low motivation	Starting or continuing a large reading task can be difficult	The learner may struggle to turn intention into sustained action.
Restlessness	Some learners need sensory or movement input	A static interface may not accommodate that need.

The hackathon opportunity is therefore not simply “make a PDF prettier.” It is to redesign the environment around the learning material so that useful engagement supports the learning objective.

2. Core Design Philosophy

The central rule in the supplied project brief is Engagement Balance: every feature must act as cognitive scaffolding that improves focus or comprehension, rather than becoming a dopamine trap that causes the learner to interact with the software instead of studying.

Four design principles

- **Function first:** visual and interactive elements must have a study purpose.
- **Reduce friction:** keep the learner close to the source material.
- **Reward learning:** tie feedback to meaningful study actions such as completion or recall.
- **Protect flow:** use guardrails, Deep Work mode and restrained feedback to prevent over-stimulation.

Retro-future visual direction

The UI direction is Cassette Futurism / precision analog hardware: warm industrial surfaces, tactile controls, crisp borders, analog meters and instrument-like panels. The brief explicitly rejects generic cyber-neon gradients, unreadable glassmorphism and decorative HUD elements that do not serve the

learner.

3. Feature System

AREA	FEATURES	PURPOSE
Reading aids	Synchronized TTS + karaoke highlighting; Bionic Reading; line status filter/mask; opacity slider; expandable	Improve reading comprehension and retention
Sensory support	Brown noise; focus beats; rain; fidget dock; breathing or eye movement control; movement options for restless	Provide controlled sensory/movement options for restless
Workspace	Dual-focus split screen; smart notes; priority tasks; flashcards; LPR, etc. switching mode	break dense material into manageable chunks
Motivation	XP; levels; streaks; milestone badges; subtle completion feedback	Provide progress without making the interface the main

4. Active Recall & Cognitive Scaffolding

Smart Pause & Test

When a learner finishes a chunk or reaches a dense section, FocusFlow can insert a non-intrusive one-question micro-quiz based on the material. Correct answers provide a small XP reward. Incorrect answers are non-punitive: the interface points back to the source sentence or paragraph so the learner can recover the information and retry.

Margin scaffolding

Complex paragraphs can be represented by short anchor labels in the margin. Selecting an anchor brings attention back to the corresponding paragraph, giving the learner a quick map of what matters without replacing the original material.

Micro-rewards

Instead of disruptive full-screen celebrations, the project uses subtle perimeter glows and small XP feedback. The intent is to preserve flow while still making progress visible.

5. Time Awareness & Executive Function

ADHD-adjusted reading time

The documented implementation estimates remaining reading time using word counts and a stated baseline of approximately 185 words per minute with a 1.25× working-memory processing buffer. The interface can display a live estimate such as “~2m 40s left (ADHD Pace).”

Break orchestration

The system tracks active reading interaction such as mouse, keyboard and scrolling activity. After a documented 25-minute continuous focus period, it can present a low-friction movement/eye-rest break with options to take a short break or snooze it.

6. Anti-Distraction Guardrails

GUARDRAIL	BEHAVIOR	DESIGN INTENT
90-second fidget cooldown	Sensory tools transition to a calm return-to-reading card after 90 seconds of fidgeting.	Keep sensory fidgeting from becoming a separate entertainment activity.
Focus Drift Detector	After an idle period or a return from another browser tab, the reading rate is lowered gently to ease aggressive interruptions.	Reading rate is lowered gently to ease aggressive interruptions.
Deep Work Mode	Shift + F hides toolbars, panels, drawers and fidget tools, leaving the document strip and controls visible.	Make the document strip and controls visible.

7. Retro-Future / Hardware UI

The visual refactor is intentionally different from generic “vibe-coded” futuristic interfaces. It uses physical metaphors to improve affordance and clarity.

- **Rack-mount dashboard:** document display, status LEDs, reading pace and session telemetry.
- **Tactile controls:** hard-edged buttons, switch-like toggles and analog-inspired dials.
- **Monitor bezel:** the reading surface is framed like a dedicated instrument display.
- **Index-card workspace:** Notes, Tasks, Cards and TL;DR appear as organized filing tabs.
- **Tape-deck TTS:** playback controls resemble a physical audio transport.
- **Pocket synthesizer fidget dock:** tactile controls for sensory tools.
- **Accessibility:** high contrast, legible body typography and keyboard operation remain more important than decoration.

8. Technical Architecture

The supplied documentation describes a lightweight, modular architecture rather than a heavy framework-based application.

FILE / MODULE	ROLE
index.html	Main application structure, Deep Work layout, margin scaffolding and time-estimate UI.
styles.css	Retro-future hardware styling, attention-pulse animations and subtle feedback.
app.js	Central orchestrator for feature modules and keyboard shortcuts.
server.py	Lightweight Python 3 HTTP server with CORS and MIME support.
active-recall.js	Micro-quiz gating and margin scaffolding.
time-awareness.js	Reading-time estimation and active-break tracking.
anti-distraction.js	Fidget cooldown, focus drift cues and Deep Work mode.
pdf-viewer.js	Document rendering enhancements and attention anchors.
chunking-mode.js	Bite-sized reading and quiz integration.

FILE / MODULE	ROLE
sensory-dock.js	Fidget tools and session countdown.
bionic-engine.js	Fixation bolding / Bionic Reading.
tts-engine.js	SpeechSynthesis and synchronized karaoke highlighting.
audio-synthesizer.js	Procedural Web Audio soundscapes.
focus-timer.js	Focus intervals and movement breaks.
dual-workspace.js	Notes, tasks, flashcards and TL;DR workspace.
gamification.js	XP, streaks, levels and badges.
storage.js	LocalStorage persistence.

9. Keyboard Interaction

SHORTCUT	ACTION
Shift + F	Toggle Minimalist Deep Work Mode
Esc	Exit Deep Work / close modal
Q	Generate an instant micro-quiz from active text
M	Trigger a movement-break card
B	Toggle Bionic Reading
R	Toggle Reading Ruler / Mask
Shift + Space	Play / pause TTS narration
→ / PageDown	Next page / chunk
← / PageUp	Previous page / chunk
Space	Step through chunking mode

10. Planned PDF-Aware AI Assistant

The next major evolution described in the project brief is a toggleable AI assistant sidebar that looks at the uploaded PDF and turns the current material into study actions.

MODE	WHAT IT SHOULD DO
Flashcards	Generate revision cards grounded in the uploaded material.
Quiz	Create questions from the current document or topic.

MODE	WHAT IT SHOULD DO
Deeper Into the Topic	Explain a concept or paragraph in more depth.
TL;DR / Summary	Condense selected or current material while preserving the learning context.
Study guidance	Help the learner decide what to review or explore next.

The same Engagement Balance principle should govern the AI sidebar: it should be collapsible, source-aware and action-oriented, rather than becoming a second feed of unrelated content.

11. End-to-End User Journey

1. **Upload:** learner provides an existing PDF/e-material.
2. **Prepare:** FocusFlow presents the material through adjustable reading aids and themes.
3. **Focus:** learner reads with TTS, Bionic Reading, ruler/mask or chunking as needed.
4. **Understand:** margin scaffolding, notes, TL;DR and micro-quizzes support comprehension.
5. **Regulate:** soundscapes, sensory tools and movement breaks can support the session.
6. **Protect:** fidget cooldowns, drift cues and Deep Work mode reduce distraction.
7. **Recall:** flashcards and quizzes reinforce the material.
8. **Progress:** XP, streaks and milestones provide restrained motivation.

12. Why FocusFlow Is Different

The project's strongest differentiator is not any one feature. It is the combination of supports around the same learning source.

TRADITIONAL TOOL	FOCUSFLOW APPROACH
PDF is mainly a document viewer.	PDF becomes the center of a focused learning workspace.
TTS is separate from reading flow.	Audio and visual highlighting are synchronized.
Notes and revision require switching tools.	Notes, tasks, flashcards and TL;DR sit beside the source.
Gamification can become an end in itself.	Rewards are tied to learning actions and balanced with guardrails.
One interface for everyone.	Themes, typography, sensory controls and focus aids can be tuned.

13. Current State & Next Development Priorities

The supplied documentation records a working prototype with the core reading, sensory, workspace, motivation, time-awareness and anti-distraction systems described above. It also records the retro-future hardware UI overhaul and a planned PDF-aware AI assistant.

Recommended implementation sequence

- Audit every visible feature button against its actual JavaScript event handler.
- Increase the physical size and visual hierarchy of core feature controls so they are noticeable at a glance.
- Add the collapsible AI assistant shell without permanently reducing reading space.
- Connect the assistant to the uploaded PDF context and expose Flashcards, Quiz, Deeper Dive and TL;DR actions.
- Test the complete keyboard shortcut map and focus behavior.
- Run usability checks specifically for distraction, readability and accidental interaction.

14. Project Thesis

FocusFlow is built around a simple thesis: the learning material does not need to become more addictive; the path to engaging with it needs to become less unnecessarily difficult.

By combining reading aids, active recall, sensory regulation, executive-function support, organization and carefully bounded motivation in one lightweight workspace, the project aims to make the learner's attention work for the subject — not for fighting the interface.

The intended outcome is not more time spent inside FocusFlow. It is more of the learner's available time spent actually learning.

Source basis: user-provided FocusFlow project documentation and supplied implementation notes.